Anique Ali

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Education

Carleton University Ottawa, ON

Bachelors of Software Engineering | Dean's Honour List, 2025-26 *Expected Graduation: June 2027*

Relevant Coursework:Algorithms and Data Structures, Operating Systems, Real Time Concurrent Systems, Object Oriented Programming, Computer Architecture, Software Engineering practice, Computer Communications

Technical Skills

Languages: Java, Python, C, SQL, JavaScript, HTML, CSS, Go, Racket
Frameworks & Libraries: React, Node.js, Flask, JUnit, FastAPI, scikit-learn, pandas, tensorflow
Tools & Platforms: Git, GitHub, Docker, Google Cloud Platform, VS Code, IntelliJ, Jupyter, Linux Terminal, Power BI, Cognos, Microsoft Excel, Microsoft PowerPoint, Microsoft Word, Azure
Concepts: SaaS, DevOps, Agile Development, Iterative Development, Test-Driven Development

Projects

Fall Detection System | *Python, Machine Learning*

- Contributed to a course project building an end-to-end fall detection system by training and evaluating an ML model in **Python** and integrating it into the application workflow.
- Developed the backend in Python (REST API) and frontend to collect user inputs, display fall-detection results, and maintain logs of detected falls for user review.

Firefighting Drone Swarm Simulator | *Java, UDP Sockets, Real-Time Systems*

- Developed a distributed simulator for a multi-drone firefighting system using **Java and UDP sockets**, where drones, a scheduler, and fire incident subsystems communicated over a network to coordinate **real-time** responses and ensure **fault tolerance**.
- Designed state machines for drones, implemented time-sensitive communication between components, and developed a **GUI** to visualize drone behavior, communication flow, and fire extinguishing progress across the **networked environment**.

Interrupt System Simulator | *C, Linux , Testing*

- Designed and implemented a **C simulator** to analyze **interrupt handling performance**, using various trace files for input testing on a **Linux** environment.
- Developed and tested the simulator in a **Linux**-based system, utilizing **system calls, process scheduling**, and file I/O mechanisms for realistic performance evaluation.

Survey Management System | *Java, Spring Boot, Spring Data JPA, HTML, JavaScript, CI/CD*

- Built a Mini-SurveyMonkey web app with an HTML and JavaScript frontend and a **Spring Boot** REST backend for creating and completing surveys.
- Persisted surveys and responses using **Spring Data JPA** repositories and improved reliability with a **CI/CD pipeline** and **integration testing** across **API** and **database** flows.

Experience

Teaching Assistant Jan 2025 - Present

Carleton University

- Guided **60 students** on complex technical concepts, offering support and solutions to technical questions, fostering **critical thinking** and **problem-solving skills**.
- Provided detailed **technical feedback** on **project proposals**, **preliminary reports**, and **final drafts**, ensuring clarity, correctness, and adherence to **technical standards**.

Program Innovation & Reporting Intern Sep 2025 - Dec 2025

Canadian Air Transport Security Authority (CATSA)

- Conducted **data analysis** by gathering, merging, and validating datasets from multiple sources, ensuring accuracy through **quality assurance** and preparation for reporting.
- Supported operational decision-making by performing **ad-hoc analyses**, conducting **user acceptance testing** for BI reports.